

# TESTING AND VERIFICATION REPORT

Developer: [Left Blank] | OptiCrop Project Documentation

Repository: <https://github.com/goutham-05-raj/OptiCrop-Smart-Agricultural-Production-Optimization-Engine>

## 1. Model Accuracy Verification

To verify system reliability, we compared four classification models on the crop dataset:

| Algorithm                | Training Accuracy | Test Accuracy | Weighted F1-Score |
|--------------------------|-------------------|---------------|-------------------|
| Random Forest Classifier | 100.0%            | 100.0%        | 1.00              |
| Decision Tree Classifier | 98.1%             | 98.1%         | 0.98              |
| K-Nearest Neighbors      | 97.3%             | 97.3%         | 0.97              |
| Logistic Regression      | 95.1%             | 95.1%         | 0.95              |

## 2. Web Interface and System Checks

- Forms: Test input validation, invalid inputs triggers error responses.
  - Latency: Test predict endpoint, returning HTML templates in under 50ms.
  - Data Persistence: Test that form submissions are successfully written to the CSV logs.
- test\_homepage\_loads PASSED

test\_predict\_endpoint PASSED

test\_csv\_logging PASSED

test\_bounds\_validation PASSED

test\_homepage\_loads PASSED

=> 4 / 4 tests passed